

STATISTICS 475: Applied Discrete Data Analysis
Final Exam
April 18, 2024

INSTRUCTIONS: **DO NOT TURN THIS PAGE UNTIL TOLD TO DO SO!**

- Answer all parts of all problems in the space provided.
- Place final answers in the blanks when they are provided.
- **Show work!** If you don't have a calculator, it is enough to write out the exact equation with the right numbers so that I could calculate the result myself.
- If you need more space (I don't think you should), indicate clearly where the problem continues. **DO NOT WRITE ON THE BACK OF ANY PAGE.**
- Use $\alpha = 0.05$ unless told otherwise

PROBLEM 1 (4 pts)

Refer to the Drink Consumption data description and the output contained in Output for Problem 1. The first half-page of output is just getting familiar with the data. The actual analysis starts in the middle of the first page.

1. **(1 pt)** Look at the results from `Obj4[1:10,]`. Which of the following are TRUE? Choose all that are true.
 - (a) **The best model according to BIC has 5 variables in it.**
 - (b) Only these 10 models were identified as having positive posterior probability of being the best model.
 - (c) There are several models with posterior probabilities of at least 10%
 - (d) **All of the 10 best models have variable 2 in them**
 - (e) None of the above are true
2. **(1 pt)** Look at the second page of output. Which variables are more likely than not to be important in the model?
age, desired, nevent, nrel, state
3. **(2 pt)** One of the main goals of the research was to determine whether increased alcohol consumption (`numall`, the response variable) might be influenced by larger numbers of negative events or negative relationship experiences (variables `nevent` and `nrel`, respectively). Use the results to try to answer this question: what information are you using, and what does it suggest about the answer?
Model-averaged parameter estimates (and CIs):
`nevent`: 0.56 (0.38, 0.83); `nrel`: 1.20 (1.05, 1.38)
Larger `nrel` does relate to higher consumption, but larger `nevent` relates to decreased consumption.

PROBLEM 2 (Bonus)

What bad thing about himself does Tom keep complaining about?

1. He is a slow thinker and a fast talker, which gets him into trouble sometimes
2. **He is an impatient perfectionist who hates mistakes but can't stand proof-reading**
3. All his clothes are either stained or wrinkled
4. His glasses are crooked, so one eye or the other is always blurry.
5. His knee makes noise while he lectures, and it distracts him.
6. He eats messy food while he works, so now his keyboard is gross.

PROBLEM 3 (4 pts)

Refer to the Drink Consumption data description and the output contained in Output for Problem 2. Note that the second page of output has been edited to all fit on one page by moving the second half of the listing (row numbers 60–160) to the right of the first half.

1. (2 pt) Refer to the `summary()` output on the fitted model.
 - (a) What part of this output can be used to assess how well the model fits the data?
Residual deviance and corresponding degrees of freedom
 - (b) Use this information and draw some tentative conclusions about the the fit of the model.
161.05/79 = 2.03. *Guidelines are $1+2\sqrt{2/df_r} = 1.31$ and $1+3\sqrt{2/df_r} = 1.48$. Since 2.03 is bigger than these thresholds, we suspect that the model is NOT a good fit.*
2. (2 pt) Look at the plot on the third page of the output. Summarize what this plot tells you about how well our model fits these data. Mention at least two specific details on the plot that help you with your description.
 1. *other than at the far right end, which we should ignore, the LOESS curve wobbles gently around 0. This suggests no curvature and therefore no bad link.*
 2. *There are several of points above 3 and even above 4, and too beyond ± 2 . This indicates overdispersion.*
 3. *The variance is relatively consistent across the plot, so the variance of the distribution is not totally wrong. [other than for the overdispersion]*

PROBLEM 4 (5 pts)

A new test for a sexually transmitted disease was developed that can get results much faster than the old test, which is known to be very reliable. The two tests were compared in an experiment at a hospital where samples from 676 people were tested using both tests. Each test reported each sample as either positive or negative, and the counts are summarized in the table below.

		New		Total
		Pos	Neg	
Old	Pos	190	7	197
	Neg	15	464	479
Total		205	471	676

In all of the parts below, you should write out the formula to answer each question using numbers from the table. You may complete the calculation, but this is not necessary.

1. **(1 pt)** Estimate the probability that the new test gives a positive result, given that the old test does.

Answer: 190/197

2. **(1 pt)** Estimate the probability that both tests give a positive result.

Answer: 190/676

3. **(1 pt)** Find a 95% confidence interval for the probability in part (b)
Agresti-Coull or Score are preferred, although Wald would be similar because of the large sample.

AC: $\tilde{\pi} = 192/680 = 0.2824$; CI is $192/680 \pm 1.96\sqrt{(192/680)(1 - 192/680)/680}$

Score: $\tilde{\pi} = (190 + 1.96^2/2)/(676 + 1.96^2) = 0.2807$

$\tilde{\pi} \pm \frac{1.96\sqrt{676}}{676+1.96^2} \sqrt{(190/676)(1 - 190/676) + \frac{1.96^2}{4*676}}$

Wald: $190/676 \pm 1.96\sqrt{(190/676)(1 - 190/676)/676}$

4. **(1 pt)** Find the odds ratio representing the odds that the new test gives a positive result when the old test is positive vs. when the old test is negative. Write it as a ratio of ratios.

Answer: (190/7)/(15/464)

5. **(1 pt)** Provide the test statistic for testing whether the two diagnostic tests have the same probability of finding positives.

This is a paired data problem, so use McNemar

Answer: $M = (15 - 7)^2/(15 + 7)$

PROBLEM 5 (13 pts)

Refer to the output contained in Output for Problem 5. This is a continuation of the analysis that we performed on Midterm 2 relating to fiber-enriched crackers. Rereading the description below is not essential for this problem.

An experiment was conducted in which people were given crackers made with different fiber supplements: bran, gum, both, or none. Later they were asked whether they subsequently experienced any digestive bloating, with possible responses none, low, medium, and high. *The researchers' main question is (1) whether the source of fiber is related to the amount of digestive bloating a person experiences, and if so, (2) which fiber sources might lead to increased bloating.*

The **fiber** factor actually consists of combinations of two sources of fiber, bran and gum. On the first page of the output, I have created 2-level factors for each of these two sources, with levels **yes** when that source is present and **no** when it is absent. There are no questions to answer about this page.

1. **(2 pts)** The second page of the output contains 4 model fits. For each model, indicate how many parameters are estimated by the model. Show some kind of arithmetic to demonstrate where the answers came from.

(a) **fib2:** $1+3+3+9=16$ (Intercept, main effects, interaction)

(b) **fib3:** $1+1+1+3+1+3+3+3=16$ (Intercept, 3 main effects, 3 2-ways, 3-way)

(c) **fib4:** $1+1+1+3+1+3+3=13$ (Intercept, 3 main effects, 3 2-ways)

(d) **fib5:** $1+1+1+3=6$ (Intercept, 3 main effects)

2. **(1 pt)** Which of these models is the homogeneous association model?

Answer: fib3

3. **(1 pt)** Does the relationship between one fiber source and the level of bloating differ depending on whether the other fiber source is present? Cite evidence and draw conclusions.

No (not significantly). Test for 3-way interaction has test stat 0.74 and p-value 0.86.

4. **(1 pts)** Do the mean ratios between any two levels of bloating differ for crackers *with* bran compared to *without* bran? Cite evidence and draw conclusions.

*No (not significantly). Test for **bran:bloat** interaction has test stat 1.57 and p-value 0.67.*

5. (1 pt) Starting with `fib4.1`, we study the relationship between gum fiber and level of bloating. Look only at the results from the `ratio` column created from object `fib4.2`. What do these numbers suggest about the relationship?

Bloating tends to be more severe with gum fiber than without.

6. (1 pt) Do the confidence intervals created from object `fib4.2` suggest that the observations you made in Part 5 represent real relationships? Explain.

Partly, but mostly no. Only one of the CIs excludes 1 (high vs. none). All others contain 1, suggesting no evidence of a real difference of bloating severity with or without gum

7. (2 pts) The model created in `fib4` can be represented in abbreviated form as

$$\log(\mu_{ijk}) = \beta_0 + \beta_i^{Br} + \beta_j^G + \beta_k^{Bl} + \beta_{ij}^{BrG} + \beta_{ik}^{BrBl} + \beta_{jl}^{GBL},$$

where we use superscripts *Br* for the bran factor, *G* for the gum factor, and *Bl* for the `bloat` factor, and $i = 1, 2$; $j = 1, 2$; $k = 1, 2, 3, 4$. Use algebra to show that the odds ratio in the first line in the output from `fib4.3` is a function of only one parameter.

For gum, yes is level 2 and no is level 1. For bloat, low is level 2 and none is level 1

$$\begin{aligned} \log(OR_{21,21}^{G:Bl}) &= \log(\mu_{i22}) + \log(\mu_{i11}) - \log(\mu_{i12}) - \log(\mu_{i21}) \\ &= (\beta_0 + \beta_i^{Br} + \beta_2^G + \beta_2^{Bl} + \beta_{i2}^{BrG} + \beta_{i2}^{BrBl} + \beta_{22}^{GBL}) + (\beta_0 + \beta_i^{Br} + \beta_1^G + \beta_1^{Bl} + \beta_{i1}^{BrG} + \beta_{i1}^{BrBl} + \beta_{11}^{GBL}) \\ &\quad - (\beta_0 + \beta_i^{Br} + \beta_1^G + \beta_2^{Bl} + \beta_{i1}^{BrG} + \beta_{i2}^{BrBl} + \beta_{12}^{GBL}) - (\beta_0 + \beta_i^{Br} + \beta_2^G + \beta_1^{Bl} + \beta_{i2}^{BrG} + \beta_{i1}^{BrBl} + \beta_{21}^{GBL}) \\ &= \beta_{22}^{GBL} + \beta_{11}^{GBL} - \beta_{12}^{GBL} + \beta_{21}^{GBL} \end{aligned}$$

All parameters with a 1 in them are set to 0 by R, so only β_{22}^{GBL} remains

8. **(3 pts)** In the table below, write out the indicator variables and their values *for each combination* of the factors for **bran** (levels **no** and **yes**), **gum** (levels **no** and **yes**), and **bloat** (levels **none**, **low**, **medium**, and **high**). List the factor labels in the order given. There are more rows than you should need.

X bran	Z gum	W bloat	Indicators							
			x_1	x_2	z_1	z_2	w_1	w_2	w_3	w_4
no	no	none	1	0	1	0	1	0	0	0
no	no	low	1	0	1	0	0	1	0	0
no	no	medium	1	0	1	0	0	0	1	0
no	no	high	1	0	1	0	0	0	0	1
no	yes	none	1	0	0	1	1	0	0	0
no	yes	low	1	0	0	1	0	1	0	0
no	yes	medium	1	0	0	1	0	0	1	0
no	yes	high	1	0	0	1	0	0	0	1
yes	no	none	0	1	1	0	1	0	0	0
yes	no	low	0	1	1	0	0	1	0	0
yes	no	medium	0	1	1	0	0	0	1	0
yes	no	high	0	1	1	0	0	0	0	1
yes	yes	none	0	1	0	1	1	0	0	0
yes	yes	low	0	1	0	1	0	1	0	0
yes	yes	medium	0	1	0	1	0	0	1	0
yes	yes	high	0	1	0	1	0	0	0	1

9. **(1 pt)** Use the indicators above to write out the regression form of the linear predictor that R fits when we use
`formula = count ~ bran + gum*bloat`

$$\log(\mu_{ijk}) = \beta_0 + \beta_2^{Br} x_2 + \beta_2^G z_2 + \beta_2^{Bl} w_2 + \beta_3^{Bl} w_3 + \beta_4^{Bl} w_4 + \beta_{22}^{GBl} z_2 w_2 + \beta_{23}^{GBl} z_2 w_3 + \beta_{24}^{GBl} z_2 w_4$$

PROBLEM 6 (5 pts)

Continuing Problem 5. Refer to the output contained in Output for Problem 6. In this problem we make use of the fact that the four levels of `bloat` are ordered, from least bloating ("`none`") to most ("`high`").

1. **(1 pt)** Somewhere on the first page in the output is a test of linear association between bloating severity and including `bran` as a fiber source. Report the test statistic and p-value for this test

Test Stat 0.9997 p-value 0.317

2. **(1 pt)** Somewhere on the first page in the output is a test of linear association between bloating severity and including `gum` as a fiber source. Report the test statistic and p-value for this test

Test Stat 9.59 p-value 0.0019

3. **(1 pt)** Starting with `fib7`, we study the relationship between gum fiber and level of bloating. In object `fib7.1`, what, exactly, do the quantities under the `emmean` column represent?

Estimated Log mean number of people experiencing each level of bloating with each level of gum fiber.

4. **(1 pt)** In object `fib7.3`, all of the odds ratios are estimated to be the same. What would I have to do differently if I thought that the last odds ratio should naturally be larger than the other two?

Make the score gap between medium and high larger than 1 (the other score gaps)

5. **(1 pt)** Is there some aspect of the relationship between gum fiber and bloating that is not well explained by the linear association model? Cite evidence from somewhere in the output and interpret the result.

Compare `fib4` to `fib6g`. LR test stat is 0.099, p-value is 0.95.

The association between bloating and gum fiber is reasonably well explained by the linear association model.

PROBLEM 7 (9 pts)

1. **(1 pt)** In the context of GLMs, what does the term “link function” refer to? **(Circle ONE)**
 - (a) **A function that relates the predictor variables to the response variable**
 - (b) A function that transforms the response variable to ensure it follows a specific distribution
 - (c) A function that determines the shape of the likelihood function
 - (d) A function that estimates the standard errors of the coefficients
 - (e) All of the above
 - (f) None of the above
2. **(1 pt)** In a logistic regression, if the likelihood function is maximized to estimate the parameters of a particular model, what benefit do we gain? **(Circle ONE)**
 - (a) An exact formula for each parameter estimate
 - (b) A guarantee each parameter estimate is within two standard errors of the population value
 - (c) **A guarantee that parameter estimates would vary from sample to sample in a way that can be approximated by a normal distribution if the sample is large enough**
 - (d) The smallest value of BIC among all possible subsets of that model
 - (e) All of the above
 - (f) None of the above
3. **(1 pt)** With a fair 6-sided die, what are the odds that a person rolls a 6?

Answer: $P(6)/P(\text{not } 6)=1/5$

4. **(1 pt)** Suppose that a 75% fewer people who receive a vaccine for an illness get the illness compared to those who do not. What is the numerical value of the relative risk of disease for vaccinated people to non-vaccinated people?

Answer: 0.25

5. **(1 pt)** We often typically find confidence intervals for population quantities in a GLM on the log scale first, and then exponentiate them into intervals for parameter we are interested in. Why? **(Circle ONE)**
 - (a) The formulas for estimating most population quantities are simpler in the linear predictor scale than in the original scale

- (b) The normal approximation to the sampling distribution is closer in the linear predictor scale than the original scale
 - (c) Standard errors are easier to find in the linear predictor scale than in the original scale
 - (d) **All of the above**
 - (e) None of the above
6. (1 pt) If you are calculating a confidence interval for a probability and you don't have access to R, which method should you use?

Answer: Agresti-Coull

7. (1 pt) In a game of roulette using an American wheel, the probability of “1–12” is $12/38$ when the wheel is fair. Find the probability that a player betting “1–12” every time would lose five times in a row by chance when the wheel is fair.

Answer: $\binom{5}{0}(12/38)^0(1 - 12/38)^5 = (26/38)^5$

8. Suppose that we fit a logistic regression using the probability model

$$\text{logit}(\pi) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3,$$

where x_1 is numerical and x_2 and x_3 are each indicators for separate 2-levels categorical variables (e.g., perhaps x_2 is for smoking and x_3 is for drug use).

- (a) (1 pt) If we want to find a LR confidence interval for the odds ratio corresponding to a 1-unit increase in x_1 when both factors are at their baseline levels, what coefficients would I use for `mcprofile`?

$$\begin{aligned} &(\beta_0 + \beta_1(x_1 + 1) + \beta_2 0 + \beta_3 0 + \beta_4 x_1 0 + \beta_5(x_1 + 1)0) \\ &- (\beta_0 + \beta_1 x_1 + \beta_2 0 + \beta_3 0 + \beta_4 x_1 0 + \beta_5 x_1 0) \\ &= \beta_1 \end{aligned}$$

Answer: 0, 1, 0, 0, 0, 0

- (b) (1 pt) If we want to find a LR confidence interval for the odds ratio corresponding to a 1-unit increase in x_1 when $x_2 = 0$ and $x_3 = 1$, what coefficients would I use for `mcprofile`?

$$\begin{aligned} &(\beta_0 + \beta_1(x_1 + 1) + \beta_2 0 + \beta_3 1 + \beta_4 x_1 0 + \beta_5(x_1 + 1)1) \\ &- (\beta_0 + \beta_1 x_1 + \beta_2 0 + \beta_3 1 + \beta_4 x_1 0 + \beta_5 x_1 1) \\ &= \beta_1 + \beta_5 \end{aligned}$$

Answer: 0, 1, 0, 0, 0, 1

PROBLEM 8 (3 pts)

Which distributional model is best for modeling data from the following situations, Binomial, Multinomial, or Poisson? (Circle one for each question.)

1. A traffic engineer is monitoring the number of accidents occurring daily at a busy intersection over the course of a month to assess the effectiveness of recent safety measures implemented in the area. They will compare these counts to those from a month prior to the changes. How should I model these counts?

BINOMIAL

MULTINOMIAL

POISSON

2. In NHL hockey, a team either wins, loses in overtime, or loses without overtime. For each team we have a count of the number of times each of these outcomes happened. How should I model these counts?

BINOMIAL

MULTINOMIAL

POISSON

3. A restaurant has a 5-item sampler on its menu. For a month, the owner asks each customer who orders it which item was their favourite and totals the results. How should I model these counts?

BINOMIAL

MULTINOMIAL

POISSON

4. A biologist is studying the number of mice passing a camera on a forest trail during different hours of the day to understand their feeding behaviour. How should I model these counts?

BINOMIAL

MULTINOMIAL

POISSON

5. An economist gathers data on many companies that were started in from 2015-2017 and wants to know whether certain features of the company could be used to predict whether or not the company survived past the pandemic. How should I model these counts?

BINOMIAL

MULTINOMIAL

POISSON